



SIMATS ENGINEERING



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Course Code: DSA0404

Slot: A

Course Name: Fundamentals of Data Science

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1. A university claims that at least 80% of its students are satisfied with online classes. A survey produces a 95% confidence interval of (74%,79%) for the satisfaction proportion. What conclusion can be made about the university's claim?

Given:

- University claims that at least 80% of students are satisfied.
- Confidence level = 95%
- 95% Confidence Interval = (74%, 79%)

Hypotheses:

$$H_0: p \geq 0.80$$

$$H_1: p < 0.80$$

Analysis:

The 95% confidence interval for the satisfaction proportion is:

$$74\% < p < 79\%$$

The claimed value of 80% is outside the confidence interval. Also, the entire confidence interval is below 80%.

Decision:

Since 80% is not contained in the confidence interval, we reject H_0 .

Conclusion:

There is sufficient statistical evidence at the 5% significance level to conclude that the

proportion of students satisfied with online classes is less than 80%. Therefore, the university's claim that at least 80% of students are satisfied is not supported.

2. A mobile application developer claims that users spend an average of 30 minutes per day on the application. A sample produces a 99% confidence interval of (31.5, 35.2) minutes. Using the confidence Hypothesis Testing Using CI interval, determine whether the claim is supported.

Given:

- Claimed average usage = 30 minutes/day
- Confidence level = 99%
- Confidence Interval = (31.5, 35.2) minutes

Hypotheses:

$H_0: \mu = 30$

$H_1: \mu \neq 30$

Analysis:

The 99% confidence interval is:

$$31.5 < \mu < 35.2$$

The claimed mean of 30 minutes is not included in this interval.

Decision:

Since 30 is outside the confidence interval, we reject H_0 .

Conclusion:

There is sufficient statistical evidence that the average time users spend on the application is different from 30 minutes per day. Therefore, the developer's claim of an average of 30 minutes is not supported.

3. A credit-card company claims that its fraud detection system identifies 95% of fraudulent transactions. After testing the system on a sample, the calculated p-value is 0.02. At a 5% significance level, what conclusion should the data scientist make?

Given:

- **Claimed fraud detection rate = 95%**
- **p-value = 0.02**
- **Significance level, $\alpha = 0.05$**

Hypotheses:

$H_0: p = 0.95$

$H_1: p \neq 0.95$

Analysis:

Compare the p-value with the significance level:

p-value = 0.02

$\alpha = 0.05$

Since $0.02 < 0.05$, the p-value is less than the significance level.

Decision:

Therefore, we reject H_0 .

Conclusion:

There is statistically significant evidence against the company's claim that the fraud detection system identifies 95% of fraudulent transactions. The data suggests that the actual detection rate is significantly different from 95%.

4. An e-commerce company introduces a new recommendation algorithm. The existing algorithm has a conversion rate of 12%. After implementation, the observed conversion rate is 15%. The hypothesis test produces a p-value of 0.03. At $\alpha = 0.05$, determine whether the new algorithm significantly improves conversion.

Given:

- Existing conversion rate = 12%
- New conversion rate = 15%

- p-value = 0.03
- Significance level, $\alpha = 0.05$

Hypotheses:

$H_0: p \leq 0.12$

$H_1: p > 0.12$

Analysis:

The observed conversion rate increased from 12% to 15%.

p-value = 0.03

$\alpha = 0.05$

Since $0.03 < 0.05$, the result is statistically significant.

Decision:

We reject H_0 .

Conclusion:

There is sufficient statistical evidence that the new recommendation algorithm significantly improves the conversion rate compared with the existing 12% rate.

5. A hospital introduces an AI-based patient scheduling system. The hospital wants to determine whether it reduces average waiting time. A hypothesis test produces a p-value of 0.08. Using a significance level of 5%, should the hospital conclude that the new system reduces waiting time?

Given:

- p-value = 0.08
- Significance level, $\alpha = 0.05$

Hypotheses:

H_0 : The new system does not reduce waiting time.

H_1 : The new system reduces waiting time.

Analysis:

Compare the p-value with the significance level:

p-value = 0.08

$\alpha = 0.05$

Since $0.08 > 0.05$, the p-value is greater than the significance level.

Decision:

Therefore, we fail to reject H_0 .

Conclusion:

There is insufficient statistical evidence at the 5% significance level to conclude that the AI-based patient scheduling system reduces average waiting time. Hence, the hospital cannot claim a significant reduction in waiting time based on this test.